

Responsive Web Design

What is Responsive Web Design?

Responsive Web design is the approach that suggests that design and development should respond to the user's behaviour and environment based on screen size, platform and orientation.

What is Responsive Web Design?

- practice consists of a mix of flexible grids and layouts, images and an intelligent use of CSS media queries
- When user switches from their laptop to iPad, the website should automatically switch to accommodate for resolution, image size and scripting abilities
- the website should not block the user's access to the page. In other words, the website should have the technology to automatically respond to the user's preferences
- This would eliminate the need for a different design and development phase for each new gadget on the market.

Adaptive v/s Responsive Design

	Adaptive	Responsive
Layout	Need to create different UI for various types devices.	the layout is decided by the site as per requirement of user's device resolution
Load Time	Adaptive designs generally load faster than responsive ones. This is because adaptive design only transfers necessary assets specific to each device	Took 2-3 time more time than adaptive design as it has one common set of UI for all kinds of devices.
Difficulty	adaptive designs are more difficult to build because you'll need different layouts for different devices	responsive designs only require a single layout that some argue is easier to implement
Flexibility	Adaptive design is considered less flexible because a new device with a screen size you didn't plan for could break your layout.	Responsive sites are flexible enough to work well on their own by default, even if there's a new device or screen size
SEO Friendly	Adaptive design can be challenging to SEO.	Google recommends and rewards sites that use responsive design. A mobile-friendly website ranks higher on search engine results pages.

Advantages of Responsive Web Designing

More mobile traffic

Faster mobile development at lower costs

Lower maintenance needs

Lower bounce rates

Higher conversion rates

Easier analytics reporting

Improved SEO

Improved online browsing experience

Improved offline browsing experience

Difference between Multiple Devices



Viewport

- The **viewport** is the user's visible area of a web page.
- The viewport varies with the device, and will be smaller on a mobile phone than on a computer screen.
- Before tablets and mobile phones, web pages were designed only for computer screens, and it was common for web pages to have a static design and a fixed size.

Setting the Viewport

- The **viewport** can be set through **<meta>** tag in HTML5

```
<head>  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

- A **<meta> viewport** element gives the browser instructions on how to control the page's dimensions and scaling.
- The **width=device-width** part sets the width of the page to follow the screen-width of the device (which will vary depending on the device).
- The **initial-scale=1.0** part sets the initial zoom level when the page is first loaded by the browser.

Setting the Viewport



Without Viewport



With Viewport

Media Query

- The `@media` rule is used in media queries to apply different styles for different media types/devices.
- Media queries can be used to check many things, such as:
 - width and height of the viewport
 - width and height of the device
 - orientation (is the tablet/phone in landscape or portrait mode?)
 - resolution

Media Query

- Syntax:

@media

not|only *mediatype* and (*mediafeature* and|or|not *mediafeature*)

{

CSS-Code;

}

mediatype : *all* | *print* | *screen* | *speech*

Mediafeature: *min-width* | *max-width* |

Media Query as External Stylesheet

```
<link rel="stylesheet" media="screen and (min-width: 900px)" href="widescreen.css">  
<link rel="stylesheet" media="screen and (max-width: 600px)" href="smallscreen.css">  
....
```

Typical Break Points

/ Extra small devices (phones, 600px and down) */*

@media only screen and (max-width: 600px) {...}

/ Small devices (portrait tablets and large phones, 600px and up) */*

@media only screen and (min-width: 600px) {...}

/ Medium devices (landscape tablets, 768px and up) */*

@media only screen and (min-width: 768px) {...}

/ Large devices (laptops/desktops, 992px and up) */*

@media only screen and (min-width: 992px) {...}

/ Extra large devices (large laptops and desktops, 1200px and up) */*

@media only screen and (min-width: 1200px) {...}